**25900342** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) Regular EXAMINATION October 2025

## Third SEMESTER

## Discipline Specific Core Course (DSC) - MG3DSCECO201 - QUANTITATIVE ECONOMICS - I

(2024 ADMISSION ONWARDS)

**Duration: 2 Hours** **Maximum Marks: 70**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Multiple Choice Question

Answer all questions

Each question carries **1** marks

1. If A has r elements and B has s elements, then the cartesian product A × B has [K] / [CO1]

   a). r + s elements b). r × s elements

   c). r – s elements d). $r^2 + s^2$ elements

2. The sample space of tossing 3 coins has [K] / [CO2]

   a). 4 outcomes b). 6 outcomes

   c). 8 outcomes d). 12 outcomes

3. Linear consumption function is [U] / [CO1]

   a). Qs=f(x) b). Qd=a-bP

   c). Qd=f(y) d). C=a+bY

4. If $y = x^4$, then $\frac{d^2y}{dx^2}$ [A] / [CO3]

a). $3x^2$ b). $12x^2$

c). $12x$ d). $6x$

5. Supply function is [U] / [CO1]

a). Qs=f(p) b). Qd=f(y)

c). Qs=f(x) d). Qd=f(I)

6. [A] / [CO2]

A consumer's utility function is $U(x) = 10x - x^2$. The level of consumption x that maximises utility is

a). 2 b). 5

c). 10 d). 0

7. A function is a relation in which [U] / [CO1]

a). Every element of the domain has exactly one image in the range b). One element of the domain can have many images

c). Many elements of the domain cannot have any image d). None of the above

8. [An] / [CO1]

Find lim x→2 $\frac{x^2-1}{x-1}$

a). 0 b). 1

c). 2 d). 3

9. [A] / [CO1]

*Find* $A+B$ *if its exists given* $A = \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ -4 & 5 \end{bmatrix}$ and B = $\begin{bmatrix} 5 & -4 \\ -2 & 1 \\ 0 & 3 \end{bmatrix}$

a). A + B = $\begin{bmatrix} 6 & -6 \\ -2 & 4 \\ -4 & 8 \end{bmatrix}$ b). A + B = $\begin{bmatrix} 6 & -6 \\ 4 & -2 \\ 8 & -4 \end{bmatrix}$

c). A + B = $\begin{bmatrix} -4 & 2 \\ 2 & 2 \\ -4 & 2 \end{bmatrix}$ d). A + B = $\begin{bmatrix} 2 & -4 \\ 2 & 2 \\ -4 & 2 \end{bmatrix}$

10. What is the derivative of f(x)+ g(x) with respect to x? [U] / [CO2]

a). f(x) + g(x)  b). f'(x) . g'(x)

c). f'(x) + g'(x)  d). f(x) . g(x)

**[1x10 = 10]**

## Part B

Short Answer

Answer any **10** questions

Each question carries **3** marks

11. A = { 1,2,3,4} B = { 1,3,5,7} C = {1,5,6,8 } Find [U] / [CO1]
    a) *A U (B U C)*
    *b)* (A U B) ∩ C

12. If $ln_a(2) = x, ln_a(3) = y$, express $ln_a(6) = ?$ [U] / [CO1]
    use Identity $\ln_a$ (A x B) = $\ln_a$ (A) + $\ln_a$ (B)

13. Distinguish between square matrix and rectangular matrix. [U] / [CO1]

14. [U] / [CO2]
    Find the minimum value of f(x).
    $f(x) = 3x^2 - 12x + 7$

15. What is a simple event? [K] / [CO2]

16. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, verify that $(A^T)^T = A$ . [A] / [CO4]

17. If revenue is $R = 100Q - 2Q^2$ , find marginal revenue. [U] / [CO2]

18. Illustrate addition theorem for mutually exclusive events. [U] / [CO2]

19. Solve the system of equations using Cramer's rule: [U] / [CO5]
    2x-y = 1
    7x-2y = -7

20. Find the inverse of (i) $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$ [U] / [CO4]

    (ii) $B = \begin{bmatrix} 5 & 7 \\ -2 & 3 \end{bmatrix}$

21. Find f'(x). [U] / [CO3]
    $f(x) = (3x^2 + 2)^4$

22. Find A - B and B - A . [U] / [CO1]

$A = \begin{bmatrix} 5 & 2 \\ 3 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 6 \\ 4 & 2 \end{bmatrix}.$

23. Define revenue function. [U] / [CO1]

24. State the addition theorem of probability. [K] / [CO2]

25. Define the degree of an equation. Give an example. [U] / [CO1]

**[3x10 = 30]**

## Part C

Short Essay

Answer any **5** questions

Each question carries **6** marks

26. [U] / [CO1]

Determine whether the function $f(x) = \frac{x-1}{x^2-1}$ is continuous at x=1. If not, explain why.

27. The consumption function is C=C+cY, and the investment function is I=I*. If C=40, c=0.7, and I*=80. Then fit an equation for the equilibrium level of income and estimate it. [U] / [CO5]

28. Calculate A×B and B×A [A] / [CO4]

A = $\begin{bmatrix} 4 & -2 & 6 \\ -5 & 7 & 3 \\ 6 & 4 & 6 \end{bmatrix}$ B = $\begin{bmatrix} 3 & -1 & 9 \\ -2 & -8 & -7 \\ 6 & 7 & -2 \end{bmatrix}$

29. Find the derivative of the function [U] / [CO1]

$a. y = x^6 + 4x^3 - 7x + 2$

$b. y = x^5 - 3x^4 + 2x^2 - x + 1$

$c. y = x^9 - 4x^7 + 5x^3 - x + 6$

30. Differentiate the following. [A] / [CO1]

1. $y = (x^2 + 3x)(x^3 + 2)$
2. $y = \frac{x^2+1}{x}$
3. $y = \frac{3x-5}{x+2}$
4. $y = (2x^2 + 1)^{-4}$
5. $y = \log(x^2 + 4)$
6. $y = 7e^{3x}$

31. Define random experiment, trial, and outcome. Give one example each. [K] / [CO2]

32. Find the determinant: [U] / [CO1]

A= $\begin{bmatrix} 1 & 0 & 4 \\ 3 & 5 & -1 \\ 0 & 1 & 2 \end{bmatrix}$

33. Explain the various total and marginal concepts in economics with examples from cost, revenue, and utility analysis. Also show how marginal values are derived from total functions and why marginal concepts are critical for decision-making in economics? [U] / [CO1]

34. Write the formula for conditional probability when both A and B are dependent events. [K] / [CO2]

35. Find adj(B). [U] / [CO1]

$B = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 4 & -1 \\ 5 & 2 & 2 \end{bmatrix}$

**[6x5 = 30]**